Smart Badge System for Predicting Engagement in Small Group Environments

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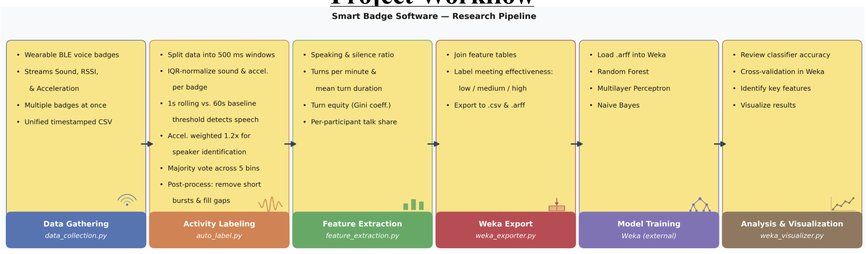
Abstract

- Developed a **BLE-based wearable badge system** to analyze meeting dynamics without recording audio, **ensuring privacy.**
- Enables **real-time multi-user sensing** and extracts features such as speaking ratio, silence ratio, and turn-taking patterns.
- **Quantifies participation equity** using metrics like the Gini coefficient.
- Uses **machine learning to classify meeting effectiveness** (low/medium/high) with up to 93% accuracy.
- Reveals that **speaking imbalance and excessive silence are strongly associated with lower meeting quality**, enabling real-time feedback for improved collaboration.

Technical Description

- Each BLE wearable badge **streams sound level, acceleration, and RSSI data to a central host**, with device whitelisting to ensure reliable participant-specific data.
- Raw **sensor signals are segmented into 20-second windows**, and features are computed from each segment.
- Speech activity is **automatically labeled using heuristic sound thresholds** and cross-badge peak comparison within 500 ms time bins.
- Extracted features include **speaking ratio, silence ratio, turns per minute, and Gini coefficient** for participation equity.
- Multiple machine learning models are trained using self-reported labels to classify meeting effectiveness, **achieving up to 93% accuracy.**

Project Workflow

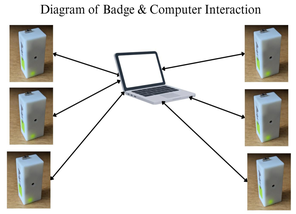


Existing Research

STUDY	APPROACH	ACCURACY / PERFORMANCE	LIMITATIONS
MIT Sociometric Badges	Wearable (audio + motion + IR + Bluetooth)	~87% (face-to-face interaction detection) (MIT Media Lab)	Expensive; derived metrics vary
Open Badges (2016)	Open-source wearable badges	N/A	Requires custom assembly
Validation Study (2018)	Commercial badge validation vs ground truth	~50–65% sensitivity, ~95% specificity (arXiv)	Vendor metrics unreliable
Accelerometer Speech Detection (2012)	Speech via vibration (no audio)	Not consistently reported (lower than audio methods) (Studocu)	Reduced accuracy
Smartwatch Conversation Detection (2025)	Smartwatch (audio + motion)	82%	Needs phone pairing
Multimodal Engagement (2024)	Smart glasses (video + audio + eye tracking)	96%	Privacy + cost issues
Conversing Group Detection (2018)	Accelerometers + transfer learning	~70–85% (task-dependent)	Needs multiple users
Hybrid Meeting Engagement (2025)	Physiological wristbands (HR, EDA)	N/A	Extra sensors required

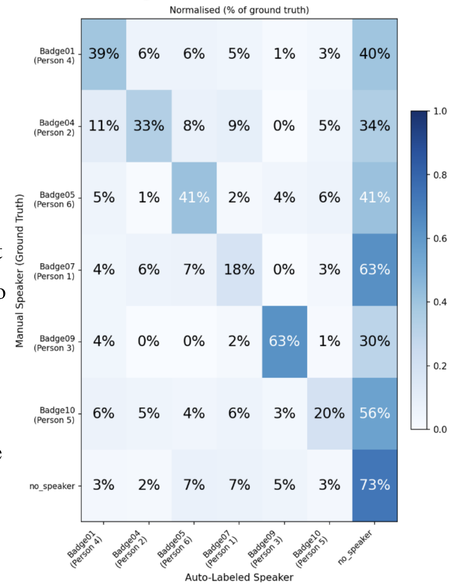
Data Collection

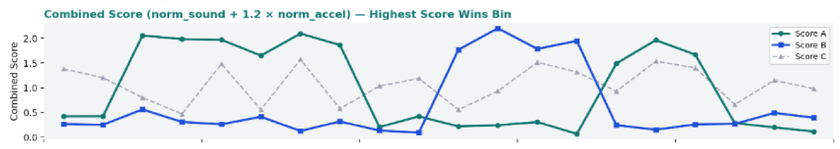
- Connect to Badges via **Bluetooth Low Energy** based on MAC
- System supports **~6-8 Badges** connected to one host
- Badge data is **pulled at 16 HZ**
- Data includes:
  - **Sound Level**
  - **Acceleration**
  - **RSSI**
- This data gets stored in a CSV file



Auto Data Labeling

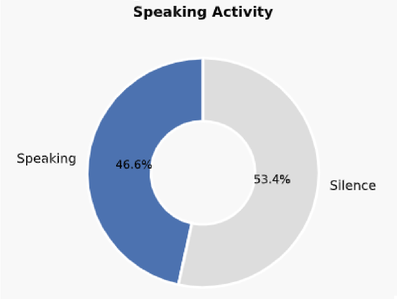
- Data is split into **500 ms time windows** for analysis
- Each badge is scored using **IQR-normalized** sound and accelerometer data
- A **rolling 1s mean vs. 60s baseline** threshold detects when speech is present
- **Accelerometer data is weighted 1.2x** to distinguish the speaking badge from bystanders
- The highest-scoring badge **wins a majority vote across 5 windows** before being labeled active
- A post-processing pass removes **short bursts, fills brief gaps, and merges nearby active periods**

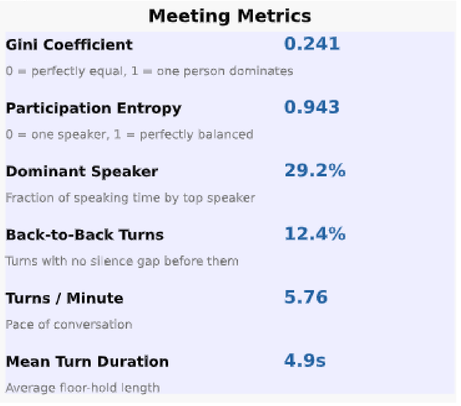




Feature Extraction

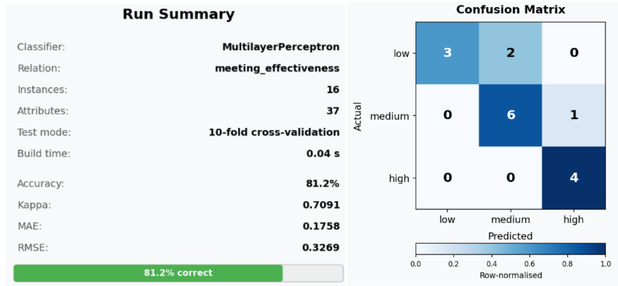
- The auto-labeled CSV is loaded, and raw samples are aggregated into **500ms bins per badge**
- The meeting is divided into **fixed time windows** (default 5 minutes) and the same features are computed per window to capture how dynamics shift over time
- Three output tables are produced:
  - one row per meeting
  - one row per participant
  - one row per time window

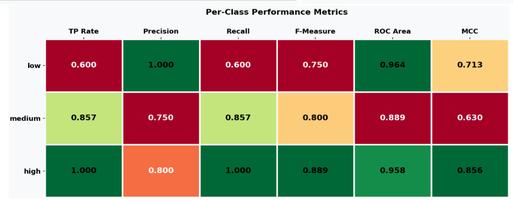




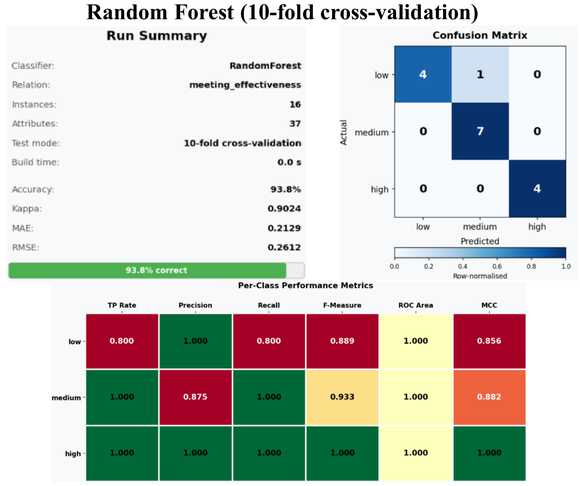
Machine Learning Results

Multilayer Perceptron (10-fold cross-validation)





Random Forest (10-fold cross-validation)



Analysis & Conclusions

- **Random Forest achieved the best result** out of the tested models, at **93.8%** (Kappa: 0.902)
- High effectiveness meetings were **classified nearly perfectly** by both models, suggesting a clear behavioral signature exists
- **Medium seems harder to classify for Rando Forest, and low was challenging for Multilayer Perceptron**
- Random Forest had **better Recall and Perception** overall
- Turn-taking dynamics and participation balance (micro turn ratio, back-to-back ratio, speaking ratio) were the **strongest predictors** of meeting quality

Future Work

- Expand dataset beyond 16 meetings to **improve generalizability** and **reduce overfitting risk**
- Test across **remote meetings, larger teams, and professional settings**
- Build a **real-time feedback dashboard** that alerts facilitators when a meeting trends toward low effectiveness
- **Increase the deployment of badges** from 6-8 to a wider amount to see if outcomes degrades or improves
- Move from WEKA to a **custom model for detection and identification**
- Attempt to get **model on-device** for faster real-time response